

SANAE IV BASE, Antarctica

WMO: 890040

Lat: 71.674S

Lon: 2.839W

Elev: 817

StdP: 91.89

Time Zone: 2.00 (E02)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-33.6	-32.0	-40.7	0.1	-30.5	-38.9	0.1	-29.8	34.1	-19.3	31.4	-24.1	10.7	120	1.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.5	-2.2	-5.1	-3.2	-5.7	-4.1	-6.2	-4.6	-2.9	-5.3	-3.7	-6.0	-4.5	4.1	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
-6.1	2.5	-5.1	-6.9	2.3	-5.7	-7.8	2.1	-6.1	2.7	-2.8	1.5	-3.6	0.5	-4.4	0.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.2	26.3	22.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-16.8	-7.3	-10.9	-15.1	-18.6	-19.7	-21.7	-23.4	-23.7	-22.9	-18.6	-12.2	-7.4		
	DBStd	6.87	2.03	2.82	3.18	4.04	4.49	4.27	4.10	4.38	4.20	3.99	2.87	2.36		
	HDD10.0	9792	536	586	777	857	920	952	1037	1046	987	888	667	540		
	HDD18.3	12834	794	819	1036	1107	1178	1202	1295	1304	1237	1146	917	798		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	10.2	7.7	8.9	10.2	9.8	12.0	11.8	10.8	11.8	10.5	10.6	9.8	8.1		
(15) Precipitation	PrecAvg	352	28	26	29	27	34	24	23	28	25	33	43	32		
	PrecMax	472	66	55	73	50	76	70	47	62	68	69	92	74		
	PrecMin	247	7	5	10	8	11	5	3	7	6	8	8	7		
	PrecStd	54	14	13	15	12	15	15	10	14	13	16	22	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.9	-2.6	-6.7	-7.8	-7.9	-10.1	-12.2	-10.2	-10.9	-9.3	-3.6	0.2		
		MCWB	-4.8	-5.4	-7.8	-9.5	-8.5	-10.9	-12.9	-10.9	-12.1	-10.6	-7.2	-3.7		
	2%	DB	-2.3	-4.9	-8.3	-9.6	-10.0	-12.3	-14.0	-13.4	-13.7	-10.6	-5.9	-2.0		
		MCWB	-5.3	-6.8	-9.6	-10.4	-10.8	-12.9	-14.7	-14.2	-15.0	-11.7	-8.3	-4.8		
	5%	DB	-3.2	-5.9	-9.4	-10.8	-11.6	-13.9	-15.7	-15.3	-15.7	-11.8	-7.1	-2.9		
		MCWB	-5.8	-7.4	-10.3	-11.6	-12.2	-14.6	-16.3	-15.9	-16.7	-12.7	-8.9	-5.4		
	10%	DB	-4.1	-7.0	-10.4	-12.3	-13.0	-15.6	-17.4	-17.2	-17.3	-13.0	-8.2	-3.9		
		MCWB	-6.2	-8.2	-11.2	-13.0	-13.5	-16.3	-18.0	-17.9	-18.2	-13.8	-9.6	-6.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.9	-5.3	-7.2	-9.0	-8.3	-11.2	-12.7	-11.1	-11.6	-10.3	-6.8	-2.9		
		MCDB	-2.2	-3.8	-6.7	-8.2	-8.0	-11.0	-12.4	-10.7	-11.2	-9.8	-4.5	-0.6		
	2%	WB	-4.7	-6.1	-9.2	-10.3	-10.6	-13.3	-14.5	-14.6	-14.9	-11.7	-7.7	-4.3		
		MCDB	-3.1	-5.0	-8.6	-9.6	-10.2	-12.7	-14.0	-13.9	-13.9	-11.1	-6.1	-2.7		
	5%	WB	-5.3	-7.3	-10.1	-11.4	-12.4	-14.9	-16.2	-16.3	-16.8	-12.7	-8.5	-5.1		
		MCDB	-3.8	-6.4	-9.6	-10.7	-12.0	-14.3	-15.6	-15.8	-15.9	-12.0	-7.2	-3.3		
	10%	WB	-5.9	-8.1	-11.2	-12.9	-13.8	-16.6	-17.9	-18.1	-18.3	-13.9	-9.3	-5.8		
		MCDB	-4.4	-7.1	-10.4	-12.2	-13.3	-16.0	-17.3	-17.5	-17.5	-13.2	-8.2	-4.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	4.5	4.3	4.6	4.6	5.1	5.1	5.0	5.0	4.8	4.5	4.3		
		MCDBR	5.9	5.7	4.9	5.4	5.0	5.9	5.9	6.1	6.3	5.6	5.6	5.6		
		MCWBR	4.5	4.8	4.3	5.0	4.9	5.4	5.6	6.1	5.5	4.9	4.6	4.1		
	5% WB	MCDBR	5.2	4.8	4.3	5.5	5.1	5.5	5.9	6.5	6.1	5.1	5.2	5.2		
		MCWBR	4.2	4.4	4.1	5.3	4.9	5.3	5.6	6.2	5.4	4.5	4.5	4.1		
(40) Clear-Sky Solar Irradiance	taub		0.220	0.206	0.180	0.122	N/A	N/A	N/A	0.120	0.164	0.185	0.204	0.221		
	taud		2.225	2.241	2.233	1.981	N/A	N/A	N/A	1.950	2.178	2.203	2.219	2.203		
	Ebn at Noon		1027	979	888	634	0	0	0	617	897	1002	1043	1042		
	Edh at Noon		111	91	63	27	0	0	0	27	64	94	111	119		
(44) All-Sky Solar Radiation	RadAvg		8.65	5.65	2.64	0.59	0.00	0.00	0.00	0.23	1.75	4.46	7.73	9.66		
	RadStd		0.19	0.17	0.09	0.03	0.00	0.00	0.00	0.01	0.08	0.14	0.26	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page